

2D Array.

↳ array inside an array.

1D → [1, 2, 3, 4]

2D (int) :-

[[1, 2, 3], [4, 5, 6], [7, 8, 9]]

	1 st col	1 st	2 nd
	↓	↓	↓
0 th row →	1	2	3
1 st →	4	5	6
2 nd →	7	8	9

1D array

int [] arr = new int [n];

2D array

int [] [] ~~matrix~~ = new int [row] [col];

new int ^{row} [2] ^{col} [3];

[[0, 0, 0], [0, 0, 0]]

	1	2	3
1	0	0	0
2	0	0	0

new int [4] [3];

[[0, 0, 0] _ _ _].

1D array

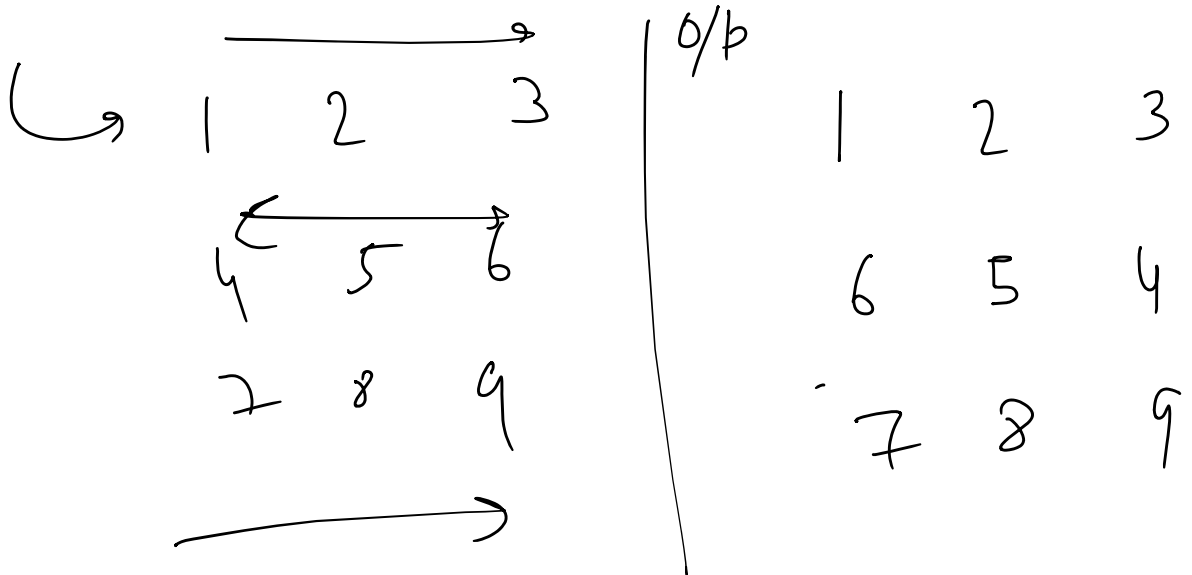
int [] arr = { 1, 2, 3 } ;

int [] [] mat = { { 1, 2, 3 }, { 4, 5, 6 }, { 7, 8, 9 } } ;

```

import java.util.Scanner;
public class Array_2d {
    public static void main(String[] args) {
        int [][] mat=new int[3][3];
        Scanner sc=new Scanner(System.in);
        for(int row=0;row<3;row++){
            for(int col=0;col<3;col++){
                mat[row][col]=sc.nextInt();
            }
        }
        System.out.println("output:");
        for(int row=0;row<3;row++){
            for(int col=0;col<3;col++){
                System.out.print(mat[row][col]+" ");
            }
            System.out.println();
        }
    }
}

```



1	2	3	4	<u>Op:-</u>	1	2	3	4
5	6	7	8		8	7	6	5
10	11	12	13		10	11	12	13
14	15	16	17		17	16	15	14

$[[1, 2, 3], [4, 5, 6], [7, 8, 9]]$

```
int [][] mat={{1,2,3},{4,5,6},{7,8,9}};

for(int i=0;i<3;i++){
    if(i%2==0){
        for(int j=0;j<3;j++){
            System.out.print(mat[i][j]+" ");
        }
    }
    else{
        for(int j=2;j>=0;j--){
            System.out.print(mat[i][j]+" ");
        }
    }
    System.out.println();
}
```

1	2	3	↓	Op	1	4	7
4	5	6	↓		2	5	8
7	8	9	↓		3	6	9

1	2	3		7	4	1
4	5	6	910 →	2	5	8
7	8	9		9	6	3